



Intention towards renewable energy investments in Malaysia: extending theory of planned behaviour

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Abstract

Renewable energy investments possess great potential for reducing the consumption of fossil fuels influenced by various determinants. This study investigates the individual investors' renewable energy investments' intention within the framework of the theory of planned behaviour (TPB) based on a survey conducted in 3 major states in Malaysia. The results indicate that one's intention to invest in renewable energy investments is influenced by attitude, subjective norm, perceived behavioural control and evaluation of regulatory framework. Risk aversion on the other hand was found to have no effect on investors' intention towards such investments. The findings also reveal that the evaluation of regulatory framework is the most important determinant. This outcome contradicts the outcomes arrived at by the previous studies that focus on investment behaviours or other types of pro-environmentally intention or behaviours. This research also investigates the indirect effects of TPB on explaining investor's intention towards renewable energy investments through the evaluation of regulatory framework. The results indicate that the investors' intention towards renewable energy investments is indirectly influenced by attitude and perceived behavioural control. Subjective norm does not have an indirect effect on investors' intention towards renewable energy investments. This study provides policymakers' important practical implications to improve renewable energy investments.

Keywords Renewable energy investments · Theory of planned behaviour (TPB) · Risk aversion · Evaluation of regulatory framework · Malaysia

Introduction

It is generally known that population growth, humans' life-style, habits and consumption patterns contribute to environmental degradation. The consumption of nature resources by human beings to fulfil their needs usually damages the environment. Fossil fuel resources such as natural gas and coal have been widely used for energy production. The depletion of fossil fuel resources and environmental degradation shift

the focus of countries such as Malaysia to other resources aiming to achieve transition to a low-carbon nation. Hence, renewable energy resources such as solar, wind, hydropower, geothermal and biomass which reduce and replace the usage of fossil fuels are introduced as a long-term source of sustainable energy.

In line with the growth of renewable energy in countries, renewable energy investments become one of the possible areas to be included in investors' portfolios. Renewable energy investments are a type of socially responsible investments, defined as investment disciplines that consider environmental, social and corporate governance criteria to generate financial returns as well as positive societal impact. However, renewable energy investments are more specific refer to the investments in stocks, bonds and funds of the companies involved in renewable energy businesses that can finance renewable energy projects which are important for a country to achieve transition to a low-carbon nation.

According to International Energy Agency (2010), the transition to a low-carbon economy will require \$7.3 trillions in investment between 2010 and 2030. However, Ajadi et al.

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(2020) reported that the unfavourable global economic condition affected the renewable energy investment level in 2020. The authors further stated that there would be great challenges in financing renewable energy projects in the future due to an increase in the government debt because most of the capital in renewable energy projects traditionally came from government, corporate bonds and large institutional investors. Thus, the development of renewable energy sectors should not fully depend on government's funds. Other funding sources such as private funds from individual investors should be considered in financing renewable energy projects.

Wunder Capital (2017) mentioned that individual investors will play an important role in the manifestation of clean energy future because they may contribute to huge amounts of capital in funding renewable energy projects underserved by the government or institutional investors. Stanley (2016) found that over 70% of active individual investors were interested in pursuing chances that integrate environmental factors into their investment decision-making causing positive environmental impact. These findings indicated that individual investors viewed renewable energy investments positively that they could serve as a great potential source in financing renewable energy projects. However, most of the literature on renewable energy often considers industrial investors as their target (Ji et al. 2017; Chaiyapa et al. 2018; Saunila et al. 2018; Li et al. 2018a; Chen et al. 2020; Wu et al. 2020) ignoring individual investors' renewable energy investments' intention which is the focus of this research.

Though there are a few studies that focus on private investors' investments in renewable energy (Aguilar and Cai 2010; Gamel et al. 2016; Salm et al. 2016), yet, most of these studies examined the effect of socioeconomic factors (Aguilar and Cai 2010; Gamel et al. 2016) and financial factors (Aguilar and Cai 2010; Gamel et al. 2016; Salm et al. 2016) on investors' investment preferences. Moreover, the previous literature focused on the impact of government's policies on renewable energy investments (Romano et al. 2017; Boute 2020; Chen et al. 2020; Egli 2020; Wu et al. 2020) without taking into consideration the impact of other factors. Besides, these studies exclude the impact of other factors emerged from psychological or behavioural theories. Thus, this study intended to create a psychological model for individual investors' intention towards renewable energy investments.

The theory of planned behaviour (TPB) has been found as a suitable social psychological model in predicting the determinants of individuals' pro-environmental actions' intention (Liao et al. 2018; Judge et al. 2019; Jiang et al. 2020) as well as individuals' investment intention in the stock market (Alleyne and Broome 2011; Phan and Zhou 2014; Palamida et al. 2018). To the best knowledge of the writers, this model has been rarely applied in the context of renewable energy investments; hence, it is applied in this research to fill up the gap.

Two additional variables, i.e., risk aversion and regulatory framework, were added by the researchers of this study to the classic TPB model by Ajzen (1991). Although most of the studies that examined the effect of risk aversion and evaluation of regulatory framework did not integrated them into any theoretical model. However, both risk aversion (Aren and Zengin 2016; Salem 2019) and evaluation of regulatory framework (Wan and Shen 2013; Li et al. 2016) were found to be a predictor of intention. Thus, the writers of this research intended to integrate risk aversion into the research model to examine its interdependence with investors' renewable energy investments' intention with the application of TPB model for the first time. Moreover, evaluation of regulatory framework was also integrated into the research model.

It was proven by Ji et al. (2017) and Niesten et al. (2018) that the existence of environmental policies as well as renewable energy subsidies affect firms' investments in renewable energy industry. Obeng-Darko (2019) also mentioned that confidence in the legal and regulatory frameworks would affect investments in the renewable energy sector. Since there is lack of empirical evidence from the individual investors' perspective, this study aims to fill the gap by integrating evaluation of regulatory framework into the current research model. According to the writers, this integration provides new knowledge and enhances the understanding of the new extended TPB model.

In recent years, many studies have investigated renewable energy context in developed countries such as Eryilmaz and Homans (2016) who studied the effect of United States renewable energy policy on wind energy investment decisions and Niesten et al. (2018) who explored firms' investment in Dutch onshore wind energy industry. On the contrary, only a few attempts have been made in developing countries especially Malaysia even though the Malaysian government made great efforts to promote the renewable energy sector. As there are cultural differences between developed and developing countries, there are definitely different factors that affect investment intention. Thus, this has brought our interest to explore on the factors that affect investment intention in developing countries. This research targets on Malaysia, a developing country in which its outcomes can be used as a benchmark for other developing and emerging countries to promote their renewable energy investments in the future.

The main objective of this study is to explore whether individual investors' renewable energy investments' intention is influenced by the attitude towards renewable energy investments, subjective norm, perceived behavioural control (PBC), risk aversion and evaluation of the regulatory framework. Moreover, this study aims to examine whether the relationship between TPB and individual investors' intentions towards renewable energy investments is mediated by evaluation of regulatory framework.

As it was previously mentioned, most of the studies on renewable energy were conducted in developed countries like Germany, United Kingdom and United States; however, the important role of individual investors was neglected by the writers of these studies. Therefore, the outcomes of this research are expected to help the related authorities such as Association of Southeast Asian Nations (ASEAN), Ministry of Science, Technology and Innovation (MOSTI), Malaysian Investment Development Authority (MIDA), Sustainable Energy Development Authority (SEDA) and Securities Commission Malaysia to obtain information about the drivers of investors' intention towards renewable energy investment. The outcomes of this research are also expected to provide better understanding for developing countries' government on the potential source to accelerate the countries' renewable energy investments as well as to achieve the sustainable development goals (SDGs).

Theoretical foundation and hypothesis development

Theoretical foundation

This study intends to investigate whether renewable energy investments' intention based on extended theory of planned behaviour is driven by psychological factors. This research investigates investors' renewable energy investments' intention based on individual investors' perspective. The original TPB model (Ajzen 1991) will serve as the main theory for the research.

A number of outstanding studies such as TPB which are widely applied to explain the intended behaviour used social psychological model to investigate human's behavioural intention. TPB (Ajzen 1991) which is an extension of theory of reasoned action (Ajzen and Fishbein 1980; Fishbein and Ajzen 1975) is a theoretical framework that explores and predicts humans' behaviour. The theory believes that an intention to act is the most powerful predictor of behaviour (Ajzen 1991). According to Ajzen (1991), intention measures the motivational factors that measure how hard people are willing to try as well as the efforts they are willing to exert to perform a certain behaviour. The intention, according to TPB, is determined by individual's attitude, subjective norm and PBC (Ajzen 1991).

Ajzen (1991) suggested that researchers should include other additional predictors to improve the predictive power of TPB in a broader context. Extended TPB is widely used in pro-environmental related studies. The incorporated variables are environmental concern (Yadav and Pathak 2016), moral norm (Tan et al. 2017), risk propensity (Alleyne and Broome 2011) and perceived policy effectiveness (Wan et al. 2014). The literature shows that most of the incorporated

variables successfully increase the explanatory power of the classic TPB model and provide a better understanding of human's behavioural intention.

Conceptual model and construct definition

Since the last decade, TPB has been regarded as one of the most popular models that have been applied to understand, explain and predict different kinds of human behaviour ranging from agriculture (Wang et al. 2019), health (Arslan and Şar 2018), information technology (Park and Ohm 2014) and service (Verma and Chandra 2018; Wang et al. 2018) to financial sectors (Li et al. 2018b; Wilson et al. 2018). In the field of pro-environmental behaviours, TPB has been found to accurately predict intentions and behaviours, such as green purchasing intention (Yadav and Pathak 2017; Grilli and Notaro 2019; Judge et al. 2019), low carbon consuming behaviour (Jiang et al. 2020), waste separation behaviour (Ma et al. 2018) and socially responsible investments intention (Adam and Shauki 2014). The previous studies reached a conclusion that TPB is a useful model for identifying different intentions and behaviours; thus, it was considered to be an important aspect for the present study.

In addition to using the TPB model in predicting individuals' actual behaviours, a number of researchers used this theory in predicting individuals' behavioural intention. Examples of these researchers are Qi and Ploeger (2019) who studied China residents' green food purchasing intention, Liao et al. (2018) who tackled the context of waste separation intention and Judge et al. (2019) who investigated sustainable house purchasing intention. The difference between behavioural intention and actual behaviour is that behavioural intention refers to the possibility that certain behaviour will take place in the future whereas actual behaviour refers to the immediate action or past action that has already occurred (Wang et al. 2018).

Ajzen (1991) mentioned that when a person has a complete control over a behavioural performance, intention alone is sufficient to predict the behaviour. Thus, it is confirmed that testing intention is sufficient in understanding the possible future actions of a person. Compared to conventional investments, renewable energy investments are still new. Consequently, potential investors are still not familiar with, so it is more suitable to investigate their possible future behaviour than their past behaviour. The "intention to invest in renewable energy investments" construct refers to individual investors' willingness to invest in the stocks, bonds and funds of the companies involved in renewable energy businesses.

Attitude is defined as the perception and overall evaluation of an individual's behaviour (Liao et al. 2018). The "attitude towards renewable energy investments" construct refers to individuals' opinions and beliefs towards renewable energy investments. According to Di Falco and Sharma (2018),

individual's beliefs and attitudes are the natural starting point of behavioural antecedents. The attitude towards behavioural intention has been proven significant in broad areas of studies including green purchasing behaviour (Trivedi et al. 2018; Choi and Kim 2019), green technologies adopting behaviour (Ojo et al. 2019) and investment intention (Warsame and Ileri 2016; Akhtar and Das 2018; Lim et al. 2018). A number of previous studies indicate that if an individual has a positive attitude towards environmental affairs or issues, he/she will act consistently in line with the behaviour based on his/her attitude. Thus, an attitude towards renewable energy investments is assumed to have a positive effect on investors' renewable energy investments' intention.

H_{1a}: attitude towards renewable energy investments has a positive influence on individual investors' intention concerning renewable energy investments.

The “subjective norm” construct in this study can be understood as the belief of an individual whether or not to perform certain behaviour based on his/her perception though it may not reflect others' real opinions. The definition is derived from Ajzen (1991) who defined the subjective norm as the effect of an individual's perceived social pressure whether to engage in the referent behaviour. Referring to Belgiawan et al. (2017), an individual is affected by the perceptions of other expectations and consequently behaves in line with them. A subjective norm was proved to have a significant impact on one's behavioural intention in collectivistic countries, such as China, India and Singapore (Sreen et al. 2018; Fan et al. 2019). Palamida et al. (2018) mentioned that Malaysia is a collectivistic country where people are more likely to be guided by external expectations. Hence, based on TPB, a subjective norm which refers to the perceived social pressure engaged in renewable energy investments has a significant impact on investor's renewable energy investments' intention.

H_{1b}: subjective norm has a positive influence on individual investors' intention towards renewable energy investments.

Perceived behavioural control (PBC) is the third factor that forms the TPB model. PBC is defined as the perceived ease or difficulty to perform a particular behaviour (Ajzen 1991). The “PBC” construct in this study refers to the control and self-efficacy involved in the renewable energy investments associated with a person's belief towards his or her controllability and capabilities to perform the referent behaviour. PBC is commonly assumed to be affected by resources, knowledge and obstacles that may either help or prevent one from performing the referent behaviour. The association of PBC and behavioural intention was confirmed by Phan and Zhou (2014), Al Mamun et al. (2018) and Toni et al. (2018), as well

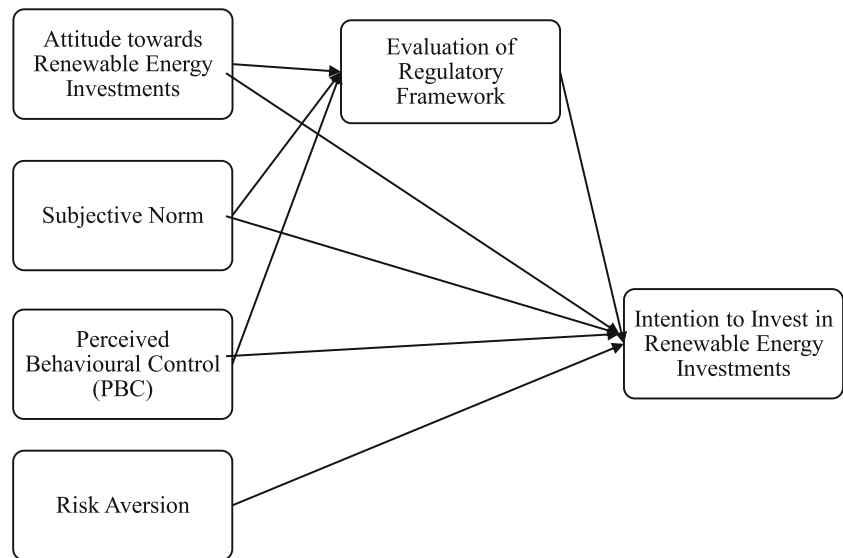
as Kaffashi and Shamsudin (2019). Based on the TPB model, investors' confidence in their ability and controllability to invest in renewable energy investments is assumed to affect their investment intention. Thus, the following hypothesis is formed.

H_{1c}: perceived behavioural control has a positive influence on individual investors' intention towards renewable energy investments.

Investors are very sensitive to risk caused by their perception and fear of losing in the investment based on the degree of risk involved in that particular investment instrument (Alleyne and Broome 2011). According to Douglas and Wildavsky (1983), risk aversion refers to an individual's attitude or tendency to avoid risk. Hu et al. (2018) mentioned that risk leads investors to invest in stock products rather than others. In other words, when investors anticipate risk in any investment in stock products, their intention to invest dwindles. Furthermore, Curtin et al. (2019) mentioned that risk aversion is one of the key barriers of investment. Gamel et al. (2017), Klein and Deissenroth (2017) and Edwards (2018) found this negative relationship in the context of energy-related investments. Ballesterio et al. (2012) found out that portfolios which have more sustainability features involve more financial risk than those having less sustainability features. Renewable energy investments require both high financing cost and a long time for evolution. Thus, risk aversion serves as one of the new factors incorporated in the extended TPB model implemented in the current study. It is expected that risk aversion will negatively affect investors' renewable energy investments' intention.

H_{1d}: risk aversion has a negative influence on individual investors' intention towards renewable energy investments.

The construct “evaluation of regulatory framework” in the current research refers to an individual's overall evaluation and opinion towards the renewable energy sector's legal and regulatory framework. The main aim of regulation is public protection (Uche 2000). However, the power of regulation depends on the willingness of society to comply with the law. Wan and Shen (2013) mentioned that policy measure acts as a motivational device may change one's behaviour. The authors also revealed that one may have stronger intention to act if he/she perceives stronger and more effective regulations. This relationship was confirmed by Gamel et al. (2017), Ozorhon et al. (2018) and Drivas et al. (2019). Similarly, according to Liu et al. (2019) in the period of 2000–2015, government engagement, stability and reliability increased the market actors' confidence. Consequently, it enhanced the willingness of market players to get involved in the renewable

Fig. 1 Research framework

energy sector leading to the growth of that sector. Hence, evaluation of the regulatory framework is integrated into the extended TPB model of this study. Based on the literature, the evaluation of regulatory framework is expected to have a positive impact on investors' renewable energy investments' intention.

H₂: evaluation of regulatory framework has a positive impact on individual investors' intention towards renewable energy investments.

The TPB constructs, i.e., attitude (Chen et al. 2016; Wan et al. 2018), subjective norm (Mauder et al. 1996; Chen et al. 2016; Moon et al. 2020; Wang et al. 2021) and PBC (Chen et al. 2016) were used in predicting individuals' policies perception while individuals' evaluation of the regulatory framework was found as a predictor of their behavioural intention. However, to the best knowledge of the writers of this study, there is no research that has been conducted to investigate the relationship between TPB constructs and renewable energy investment intention through evaluation of regulatory framework. Therefore, the writers of this study intended to examine the potential mediation effect of evaluation of regulatory framework on the relationship between the TPB constructs and renewable energy investments' intention. The mediating effect implies that the constructs, i.e., attitude towards renewable energy investments, subjective norm and PBC are indirectly associated with the individual investors' intention towards renewable energy investments via their evaluation of the regulatory framework. The research framework is depicted in Figure 1 below.

H_{3a}: attitude towards renewable energy investments has a positive impact on individual investors' evaluation of the regulatory framework.

H_{3b}: subjective norm has a positive impact on individuals' evaluation of regulatory framework.

H_{3b}: perceived behavioural control has a positive impact on individuals' evaluation of regulatory framework.

H_{4a}: the relationship between attitude towards renewable energy investments and individual investors' intention towards renewable energy investments is mediated by evaluation of regulatory framework.

H_{4b}: the relationship between subjective norm and individual investors' intention towards renewable energy investments is mediated by evaluation of regulatory framework.

H_{4c}: the relationship between perceived behavioural control and individual investors' intention towards renewable energy investments is mediated by evaluation of regulatory framework.

Methods

Population and sample

Individual investors in Johor, Kuala Lumpur and Penang, Malaysia are the population of this study. Independent investors, i.e., individuals who manage their own funds to achieve financial goals, are the unit of analysis for this research. Those over 18 in age were considered as the target population because age 18 is the minimum requirement for an individual to register the Central Depository System account in Malaysia. According to Bank Negara Malaysia (2013), 'No person who has access to any document or information relating to the affairs or account of any customer of a financial institution

shall disclose to another person any document or information relating to the affairs or account of any customer of the financial institution.’ This means that the disclosure of investors’ information is not allowed. For this reason, this research does not have a sampling frame; hence, non-probability sampling is applied.

The current study employed purposive sampling in which individual investors can be identified by using a filtering question asking whether the respondent has any investment experience for the sake of accuracy. The sample size of this research is 300. The recommended minimum sample size for this research is 92 as generated by using G*power; a programme to calculate the required sample size suggested by Tabachnick and Fidell (2013) and Hair et al. (2017). However, data analysis centres on 204 samples after filtering up the non-investors’ responses. According to Hair et al. (2017), a minimum sample size can ensure the robustness of results and generalizability of the model. Thus, the final sample size for the current study is 204.

Data collection

This research applied positivism approach and quantitative methodology. The research data for this study were gathered by using a structured questionnaire which was developed based on previous studies to collect primary data from the targeted respondents. The content validity was verified by a few professionals to ensure the adequacy of the items that the questionnaire contains. After receiving their comments, some items were amended. A pre-test assessment was then carried out on 30 participants before the distribution of the questionnaire to identify problems related to wording to ensure respondents’ understandability. Data collection was carried out between January and March 2020 in Johor, Kuala Lumpur and Penang, Malaysia.

Measurements

Based on the extended TPB model (Figure 1), six latent variables, namely, attitude towards renewable energy investments, subjective norm, PBC, risk aversion, evaluation of regulatory framework and intention are included in this study to invest in renewable energy investments. All the scales of measurement and indicators implemented in this study to measure the above stated construct variables were adopted/adapted from previous studies and modified for this study. Following Gamel et al. (2017), a five-point Likert scale was used to measure one’s attitude towards renewable energy investments, subjective norm, PBC, risk aversion and evaluation of regulatory framework. The respondents were asked to choose their option that were categorised as “strongly disagree = 1”, “disagree = 2”, “neutral = 3”, “agree = 4” or “strongly agree = 5”.

The attitude towards renewable energy investments, subjective norm, PBC and risk aversion was measured with five items taken from Gamel et al. (2017), Qi and Ploeger (2019) and Palamida et al. (2018). Evaluation of regulatory framework was measured by three items taken from Gamel et al. (2017). The intention to invest in renewable energy investments was measured by seven-point Likert scales. The respondents were asked to choose their options that were categorised as “strongly disagree = 1”, “disagree = 2”, “somehow disagree = 3”, “neutral = 4”, “somehow agree = 5”, “agree = 6” or “strongly agree = 7”. The measurement items were taken from Sivaramakrishnan et al. (2017) and Qi and Ploeger (2019). All the questions asked in the questionnaire are close-ended questions. Table 1 lists the measurements of each construct.

Data analysis

The data collected were analysed by using both Statistical Package for Social Sciences (SPSS) software and Partial Least Squares Structural Equation Modelling (PLS-SEM) software. SmartPLS version 3. SPSS software was mainly used to assess the demographic profile of the respondents. Structural equation modelling (SEM) rather than regressions was used in this research because it uses weight. According to Hair et al. (2017), SEM offers important insights to researchers on each variable’s importance in forming the composite. PLS-SEM technique was applied in the current research because of its power in analysing small samples and non-normal distributed data (Hair et al. 2017). Moreover, it is more suitable for studies that focus on theory development and variance explanation.

In accordance with Hair et al. (2017), PLS-SEM analysis contains two phases. The first phase is the measurement model assessment which ensures the validity and reliability of all the measured variables in the model. In this phase, the validity tests are conducted. The second phase is the structural model assessment that tests the significance of the proposed hypotheses and predictive power of the research model.

Results and discussion

Investors’ profile

Table 2 presents the summary of the respondents’ demographic profile statistics. There are 51% of female respondents and 49% of male respondents. Within the sample, most of the respondents are between age 31 and 40 (36.8%) and 21 and 30 (35.8%). Chinese respondents’ percentage is 48.5%, followed by Malay and Indian. In terms of education level, most of the respondents are graduates. The other respondents are represented by investors whose education level is secondary or primary. In addition, most of the respondents are from

Table 1 Constructs and measurement items

Constructs	Items	Measurement items	Label	Source
Attitude towards renewable energy investments (ATT)	5	When investing, it is important to me to invest in ethically correct companies.	ATT1	Gamel et al. (2017)
		When investing, I pay attention to the environmental friendliness of companies.	ATT2	
		When investing, I would reject higher returns which are not in favour of sustainability aspects.	ATT3	
		When investing, I examine closely for company's environmental compatibility.	ATT4	
		When investing in a company, it should be ecologically harmless.	ATT5	
Subjective norm (SN)	5	Most people who are important to me think I should invest in renewable energy investments.	SN1	Qi and Ploeger (2019)
		Most people whose opinions I value would prefer that I invest in renewable energy investments.	SN2	
		The extent of influence from my family can strongly affect my decision in renewable energy investments.	SN3	
		The extent of influence from my friends can strongly affect my decision in renewable energy investments.	SN4	
		The extent of influence from my colleagues can strongly affect my decision in renewable energy investments.	SN5	
Perceived behavioural control (PBC)	5	I feel confident about being able to engage in renewable energy investments.	PBC1	Palamida et al. (2018)
		I am able to overcome the obstacles or problems which could prevent me from engaging in renewable energy investments.	PBC2	
		Engaging in renewable energy investments is within my own control.	PBC3	
		Engaging in renewable energy investments is easy.	PBC4	
		I think I have sufficient knowledge which enables me to engage in renewable energy investments.	PBC5	
Risk aversion (RA)	5	The risk of losing money on the stock market causes mental stress.	RA1	Gamel et al. (2017)
		Stability of my investments is more important to me than the chance of a quick profit.	RA2	
		Continuity of my investments is more important to me than the chance of a quick profit.	RA3	
		Even small financial losses make me nervous.	RA4	
		I am reluctant to take risks regarding financial matters.	RA5	
Evaluation of regulatory framework (ERF)	3	Renewable energies could have to be the main power source in Malaysia's electricity market.	ERF1	Gamel et al. (2017)
		The federal government should make a clear commitment to the implementation of the energy transition.	ERF2	
		The guaranteed fixed feed for electricity from renewable energies (feed-in tariff) should be continued.	ERF3	
Intention to invest in renewable energy investments (INT)	3	I expect to invest in renewable energy-related equities (stocks/shares) and/or equity mutual funds.	INT1	Sivaramakrishnan et al. (2017)
		I would invest in renewable energy investments whenever I am given the opportunity.	INT2	
		I will make an effort to invest in renewable energy in the near future.	INT3	

the executive group ($n=69$, 33.8%) and professional group ($n=43$, 21.1%). 29 respondents are self-employed, 18 respondents are managers or senior managers and 15 respondents are non-executive. The other 30 respondents are represented by students, housewives and retiree investors.

Measurement model

The measurement model assessment was conducted to test the reliability and validity of the items and constructs before proceeding with the structural model assessment. The convergent validity of the model is verified by the factor loadings, composite reliability and average variance extracted (AVE).

Table 3 shows that all the standardised factor loadings are greater than the minimum requirement of 0.7 except for PBC4 (0.639), RA4 (0.561) and RA5 (0.611). However, these three indicators are all retained because of the contribution to content validity (Hair et al. 2017). All the composite reliability values obtained are within the range of 0.857 to 0.945 which fulfils the above criterion 0.70 (Hair et al. 2017). This confirms the indicators' internal reliability consistency. The convergent validity is further verified by AVE that tests the construct's communality (Hair et al. 2017). Referring to Table 3, the AVE values are within the above range of 0.547 to 0.851 as per the suggestion of Hair et al. (2017). Based on this, the convergent validity of these constructs is established.

The discriminant validity is assessed to test whether a construct is unique, i.e., its measures are not too highly correlated with the other constructs in the same model. The discriminant validity for this study is assessed by using the Fornell–Larcker criterion (Fornell and Larcker 1981) and HTMT criterion (Henseler et al. 2015). Fornell and Larcker (1981) criterion is examined by comparing the square root of AVE with the correlations between the constructs. Table 4 shows that all the square roots of AVE are more than the correlation coefficient between the constructs. This indicates that the discriminant validity assessed by using Fornell and Larcker (1981) criterion is established. HTMT criterion is further assessed by comparing the HTMT values obtained with the required threshold. The results in Table 5 indicate that the HTMT ratio for each of the constructs in this study is less than 0.90 (Gold et al. 2001), 0.85 (Kline 2011) and does not show a value of 1 (Henseler et al. 2015), providing further evidence of discriminant validity.

The results in Table 6 show that out of the eight paths tested in the structural model, only five paths are significant with *t*-statistics greater than critical value of 2.3263 and one path is significant with *t*-statistics greater than the critical value of 1.6449. The results indicate that six out of eight hypotheses are supported: attitude towards renewable energy investments has a positive influence on renewable energy investments intention (H_{1a} : $\beta = 0.166$, *t*-value = 2.459), subjective norm has a positive influence on renewable energy investments intention (H_{1b} : $\beta = 0.152$, *t*-value = 1.974), PBC has a positive influence on renewable energy investments intention (H_{1c} : $\beta = 0.233$, *t*-value = 3.271), evaluation of regulatory framework has a positive impact on renewable energy investments intention (H_2 : $\beta = 0.344$, *t*-value = 5.496), attitude towards renewable energy investments has a positive impact on individual investors' evaluation of regulatory framework (H_{3a} : $\beta = 0.361$, *t*-value = 6.363) and PBC has a positive impact on individual investors' evaluation of regulatory framework (H_{3c} : $\beta = 0.272$, *t*-value = 3.703).

The results in Table 6 provided no support for the fourth hypothesis (H_{1d}), risk aversion has a negative influence on renewable energy investments' intention, as the effect was non-significant (H_{1d} : $\beta = 0.037$, *t*-value = 0.603). Similarly, the results provide no support for the seventh hypothesis (H_{3b}), subjective norm has a positive impact on individual investors' evaluation of regulatory framework as the effect is not significant (H_{3b} : $\beta = 0.04$, *t*-value = 0.533). The f^2 were also being assessed. All the effect sizes among the significant constructs are greater than 0.02 as suggested by Cohen (1988).

This study also tested the indirect effects of attitude towards renewable energy investments, subjective norm and PBC on individual investors' renewable energy investments' intention through their evaluation of regulatory framework. The research results in Table 7 show that two indirect effects are supported, i.e., the relationship between attitude towards

renewable energy investments and individual investors' intention towards renewable energy investments is mediated by evaluation of regulatory framework (H_{4a}) and the relationship between PBC and individual investors' intention towards renewable energy investments is mediated by evaluation of regulatory framework (H_{4c}). However, the results indicate that the indirect effect of subjective norm on individual investors' intention towards renewable energy investments via evaluation of regulatory framework (H_{4b}) is not supported.

The results indicate that attitude towards renewable energy investments has a direct and positive effect on renewable energy investments' intention and evaluation of regulatory framework. Besides, they also illustrate that attitude has an indirect and positive influence on renewable energy investments' intention via evaluation of regulatory framework. The significant effect of attitude towards renewable energy investments is potential because the Malaysians are concerned about environmental issues as they view renewable energy investments as a type of investment that has a good effect on the environment. Malaysians are mainly formed by Malay, Chinese and Indians who hold a strong belief in their religions which are very concerned about nature preservation. Because of this, the people in Malaysia are aware of the importance of natural preservation which leads to a favourable attitude towards pro-environmentally actions. Besides, renewable energy investments' implied value is in accordance with most of the religious beliefs which strengthen nature preservation. As a result, it is reasonable for Malaysian investors to possess a positive attitude towards renewable energy investments and consequently increase their willingness to invest in them.

This study also reveals that subjective norm has a direct positive impact on individual investors' renewable energy investments' intention. This result is in line with Gerrans et al. (2018), Zhang et al. (2018) and Frey et al. (2019) who found out that there is a positive relationship between subjective norm and asset allocation decisions. The research outcomes are achieved because the Malaysian community is mostly formed by collectivistic citizens whose intentions are guided by external expectations. Thus, investors are more willing to give up their personal goals for group goals to improve the environmental quality and consequently make decisions deemed to be approved by their society.

Perceived social pressure on investors tends to affect their investment intention. The obtained results indicate that investors may increase their willingness to invest in renewable energy investments when they find out that most of the important persons around them expect them to do that in order to be considered useful for the community. Moreover, renewable energy investments in Malaysia are still at the introductory phase, so investors are expected to be unfamiliar with. Therefore, it is reasonable for investors to be persuaded by the people whom they trust to be motivated and encouraged to invest in such investments. The results indicate that

Table 2 Investors' profile

Variables		Frequency	Percentage (%)
Gender	Male	100	49.0
	Female	104	51.0
Age	Below 21	3	1.5
	21–30	73	35.8
	31–40	75	36.8
	41–50	23	11.3
	Above 50	30	14.7
Ethnic/race	Malay	77	37.7
	Chinese	99	48.5
	Indian	28	13.7
Education level	Diploma	32	15.7
	Graduate	107	52.5
	Postgraduate Degree	27	13.2
	Others	38	18.6
Occupation	Professional	43	21.1
	Manager/senior manager	18	8.8
	Executive	69	33.8
	Non-executive	15	7.4
	Self-employed	29	14.2
	Others	30	14.7
Sample size = 204			

subjective norm is insignificant with regard to the direct effect on evaluation of regulatory framework and indirect effect on

renewable energy investments via evaluation of regulatory framework. The results show that individuals' evaluation on

Table 3 Convergent validity

Construct	Items	Loadings	Composite reliability	AVE
Attitude towards renewable energy investments	ATT1	0.815	0.901	0.645
	ATT2	0.873		
	ATT3	0.758		
	ATT4	0.765		
	ATT5	0.801		
Evaluation of regulatory framework	EFR1	0.891	0.924	0.802
	ERF2	0.889		
	ERF3	0.907		
Intention to invest in renewable energy investments	INT1	0.944	0.945	0.851
	INT2	0.897		
	INT3	0.926		
Perceived behavioural control	PBC1	0.792	0.857	0.547
	PBC2	0.754		
	PBC3	0.787		
	PBC4	0.639		
	PBC5	0.707		
Risk aversion	RA1	0.753	0.875	0.594
	RA2	0.930		
	RA3	0.922		
	RA4	0.561		
	RA5	0.611		
Subjective norm	SN1	0.805	0.902	0.648
	SN2	0.779		
	SN3	0.813		
	SN4	0.844		
	SN5	0.786		

Table 4 Discriminant validity using the Fornell–Larcker criterion

	1	2	3	4	5	6
1 Attitude towards renewable energy investments	0.803					
2 Evaluation of regulatory framework	0.485	0.896				
3 Intention to invest in renewable energy investments	0.535	0.601	0.922			
4 Perceived behavioural control	0.415	0.429	0.526	0.739		
5 Risk aversion	0.588	0.486	0.402	0.211	0.771	
6 Subjective norm	0.510	0.339	0.474	0.448	0.328	0.805

Note: values on the diagonals (bolded) represent the square root of the AVE, while the off-diagonals represent the correlations between constructs.

regulatory framework is mainly affected by their own perception. Thus, their community does not affect their evaluation of regulatory framework.

The research findings reveal that the participants who perceived that they have the ability to invest in renewable energy investments showed great willingness to do that. This is similar to the findings arrived by Halder et al. (2016), Paul et al. (2016), Al Balushi et al. (2018) and Wang et al. (2019). All these studies showed that an individual's perceived ease or difficulty to perform the referent behaviour has a positive effect on their behavioural intention. In this context, investors feel that they are confident since they can have their own decisions. Moreover, investors perceived that they have high ability to invest in renewable energy investments if they want to do so. Since the investors realised that they have total control on renewable energy investments, their willingness to invest in such investments increases. The results also reveal that investors who are confident of their ability to invest in

renewable energy investments perceive regulatory framework as a tool. Consequently, it increases investors' intention to invest in renewable energy investments.

The research findings also reveal that risk aversion does not have any effect on investors' renewable energy investments' intention. This finding seems to contradict the findings arrived at by previous studies such as Aren and Zengin (2016), Gamel et al. (2017), Edwards (2018) and Taghizadeh-Hesary and Yoshino (2019) who found out that risk aversion had a negative effect on investments' intention. The results also indicate that the investors who are willing to be involved in renewable energy investments place less focus on financial returns because they might be more interested in the betterments of the society. Furthermore, the results show that since investors wish to invest consistently with their moral commitments, they value the implied moral value of renewable energy investments. In this case, the effect of risk on investments' intention may be diminished as investors prioritise non-financial

Table 5 Discriminant validity using HTMT criterion

	1	2	3	4	5	6
1 Attitude towards renewable energy investments						
2 Evaluation of regulatory framework	0.549 CI ₉₀ (0.449, 0.629)					
3 Intention to invest in renewable energy investments	0.582 CI ₉₀ (0.486, 0.668)	0.671 CI ₉₀ (0.581, 0.753)				
4 Perceived behavioural control	0.492 CI ₉₀ (0.381, 0.575)	0.494 CI ₉₀ (0.355, 0.606)	0.608 CI ₉₀ (0.477, 0.714)			
5 Risk aversion	0.643 CI ₉₀ (0.542, 0.729)	0.438 CI ₉₀ (0.346, 0.507)	0.369 CI ₉₀ (0.278, 0.458)	0.339 CI ₉₀ (0.237, 0.425)		
6 Subjective norm	0.588 CI ₉₀ (0.486, 0.68)	0.391 CI ₉₀ (0.261, 0.5)	0.519 CI ₉₀ (0.391, 0.629)	0.542 CI ₉₀ (0.402, 0.654)	0.324 CI ₉₀ (0.208, 0.449)	

Table 6 Results of partial least square (direct effect)

		Standard	Standard						
Hypothesis	Relationship	Beta	Error	<i>t</i> -value	Decision	<i>R</i> ²	<i>Q</i> ²	<i>f</i> ²	VIF
	Direct effect								
H1a	Direct effect of ATT on INT	0.166	0.068	2.459**	Supported	0.505	0.419	0.027	2.039
H1b	Direct effect of SN on INT	0.152	0.077	1.974*	Supported			0.031	1.493
H1c	Direct effect of PBC on INT	0.233	0.071	3.271**	Supported			0.075	1.461
H1d	Direct effect of RA on INT	0.037	0.062	0.603	Not Supported			0.002	1.711
H2	Direct effect of ERF on INT	0.344	0.063	5.496**	Supported			0.149	1.603
H3a	Direct effect of ATT on ERF	0.361	0.057	6.363**	Supported	0.312	0.243	0.131	1.446
H3b	Direct effect of SN on ERF	0.040	0.075	0.533	Not Supported			0.002	1.488
H3c	Direct effect of PBC on ERF	0.272	0.073	3.703**	Supported			0.081	1.319

** Significant at *p*-value <0.01, * significant at *p*-value <0.05

Note: *ATT* attitude towards renewable energy investments, *INT* intention to invest in renewable energy investments, *SN* subjective norm, *PBC* perceived behavioural control, *RA* risk aversion, *ERF* evaluation of regulatory framework

criteria due to their main objective to invest in renewable energy investments.

Finally, the research findings reveal that the evaluation of regulatory framework has a positive effect on investors' renewable energy investments' intention. This result is in line with the findings arrived at by Bauwens and Devine-Wright (2018), He et al. (2019) and Obeng-Darko (2019) who arrived at the same conclusion. The results indicate that investors value the presence of the government in the renewable energy sector. This shows that respondents value Malaysian Government reliability, so they are more willing to comply with the government policies such as net energy metering (NEM) scheme and feed-in-tariff (FiT) scheme. Investors also view regulatory framework as a helpful tool for enhancing the growth of renewable energy sector resulting in a positive

effect on their intention to invest in renewable energy investments in Malaysia.

Structural model

Referring to the methodology suggested by Hair et al. (2017), the structural model assessment was conducted to examine the model's predictive capabilities and the relationship between the constructs. The variance inflation factor (VIF) was used to evaluate the collinearity of constructs. Table 6 shows the outcomes of lateral collinearity test using attitude towards renewable energy investments, subjective norm, PBC, risk aversion and evaluation of regulatory framework as predictors of intention to invest in renewable energy investments. The table also presents the outcomes of lateral collinearity test using the TPB

Table 7 Results of partial least square (indirect effect)

Hypothesis	Relationship	Standard beta	Standard error	<i>t</i> -value	Confidence interval (BC)		Decision
					LL	UL	
	Indirect effect						
H4a	Indirect effect of ATT on INT through ERF	0.124	0.030	4.093**	0.074	0.196	Supported
H4b	Indirect effect of SN on INT through ERF	0.014	0.026	0.533	−0.04	0.065	Not Support- ed
H4c	Indirect effect of PBC on INT through ERF	0.093	0.031	2.979**	0.044	0.169	Supported

**Significant at *p*-value <0.01, *Significant at *p*-value <0.05

Note: *ATT* attitude towards renewable energy investments, *INT* intention to invest in renewable energy investments, *SN* subjective norm, *PBC* perceived behavioural control, *ERF* evaluation of regulatory framework, *BC* bias corrected, *UL* upper level, *LL* lower level

constructs as predictors of evaluation of regulatory framework. All the VIF values are between the range of 1.319 and 2.039 which is less than 5 as suggested by Hair et al. (2017). This indicates there is no collinearity problem between the predictor constructs in the structural model.

The structural model is then evaluated with a few relevant criteria: (1) the path coefficients together with the *t*-statistics and standard beta (β), (2) the coefficient of determination (R^2), (3) the effect size (f^2), and (4) the predictive relevance of the model (Q^2). The path coefficients as well as *t*-statistics were generated by carrying out a bootstrapping procedure with a resample of 5000 as suggested by Hair et al. (2017). It was used to examine the hypotheses developed for this study as shown in Figure 2. The results obtained are presented in Table 6.

According to the rule of thumb as suggested by Hair et al. (2017), the R^2 values of 0.75, 0.50 or 0.25 may be described as substantial, moderate or weak respectively. As shown in Table 6, the R^2 values for intention to invest in renewable energy investments (0.505) and evaluation of regulatory framework (0.312) indicate that the proposed extended TPB model has a moderate predictive accuracy on individual investors' intention to invest in renewable energy investments while the TPB constructs have a weak predictive accuracy on individual investors' evaluation on regulatory framework. A blindfolding procedure was used to examine the predictive relevance of the model. The results show that Q^2 values 0.419 (intention to invest in renewable energy investments) and 0.243 (evaluation of regulatory framework) indicate an acceptable level of predictive relevance (Hair et al. 2017).

As an attempt to apply the extended TPB in examining the renewable energy intention from individual investors' perspective, the results show that the extended TPB model seems to be an ideal theoretical model for predicting one's intention to invest in renewable energy investments.

Conclusions and policy implications

This study provides a quantitative assessment of individual investors' intention to invest in renewable energy investments in Malaysia. Focusing on renewable energy investments' intention, which has not been deeply investigated by previous researches, TPB was used as a theoretical foundation of this research. Risk aversion and evaluation of regulatory framework were integrated into the model on top of the original three variables.

The findings of this study show that the renewable energy investments' intention is directly determined by attitude towards renewable energy investments, subjective norm and PBC. Evaluation of regulatory framework on investors' renewable energy investments was found to be the most influential determinant on intention. The mediating analysis

revealed that individuals' evaluation of regulatory framework mediates the relationship between attitude and intention as well as PBC and intention while there is no mediating effect of the evaluation of regulatory framework on subjective norm and intention.

The study contributes to the knowledge of research on individual investors' intention towards renewable energy investments with new constructs, namely, risk aversion and evaluation of regulatory added to the research model. These constructs enhanced the predictive power of TPB model, especially in renewable energy investments. Therefore, it provided evidence on the applicability of TPB into this area. The literature will then be enriched and provides an insight in the area of renewable energy investments in Malaysia. The findings of mediating analysis indicate that the evaluation of regulatory framework which has a mediating role in the model will also add a new knowledge to this area.

Based on policymakers' perspective, the related government bodies such as Malaysian government, Sustainable Energy Development Authority and Ministry of Science, Technology and Innovation should exert more efforts in introducing more appropriate renewable energy-related policies and supporting schemes that benefit individual investors. Although many policies such as Green Technology Financing Scheme, Green Investment Tax Allowance and Green Income Tax Exemption have been introduced to promote the adoption of renewable energy in the country, most of the policies benefit industrial investors. Therefore, the government should provide tax incentives such as tax relief to individual renewable energy investors.

Renewable energy investments in most developing countries are still at the elementary stage due to the fact that most of the investors still lack awareness and information. Because of this, the government or central bank should plan or organise a special programme to raise the awareness of investors about renewable energy investments. The government could also collaborate with environmentalists in transmitting the importance of environment-related actions to potential investors in the country. Besides, the government needs to provide more credible and reliable information on renewable energy investments and regulate the spread of renewable energy investments' information to ensure the accuracy of information. These may contribute to building trust, confidence and enhance potential investors' attitude towards the actions and thereby produce more favourable responses towards the policies as well as investments.

Most of the people in developing countries such as Malaysia, Indonesia, Thailand and India live in a collectivistic culture, so they value the people who are important to them and prefer group goals. Therefore, the government may set long-term or short-term renewable energy goals. This may increase investors' willingness to invest in renewable energy investments because they expect that most of the people who

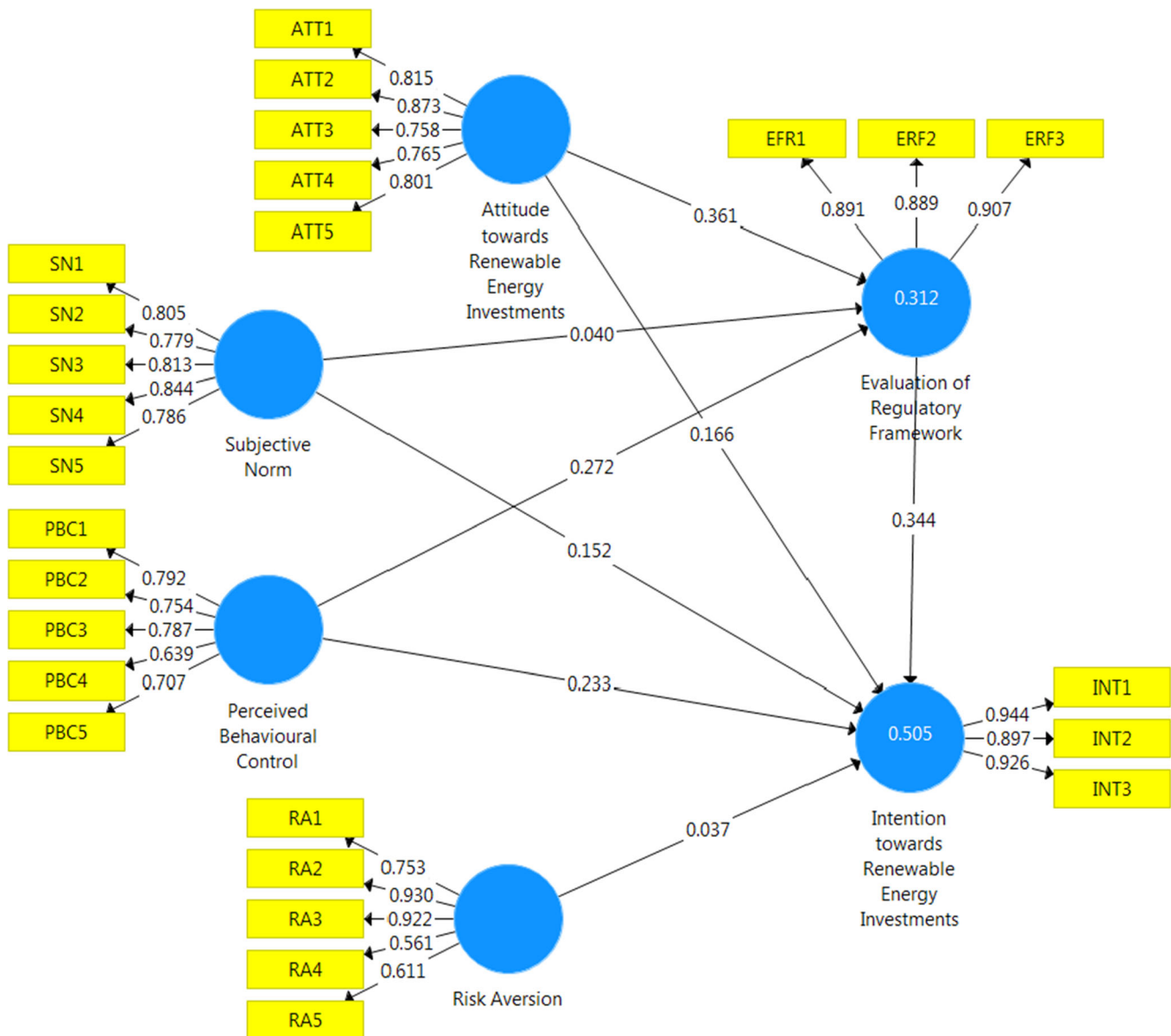


Fig. 2 Structural model for this study

are important to them support them to invest in the related investments as they want to be positively viewed in the society. Besides, the insignificant relation between risk aversion and renewable energy investments' intention implies that individual investors may have moved away from the traditional investments objective that prioritises financial criteria in their invest decision. The results indicate that the government could focus on the value of renewable energy investments in contributing to the betterments of society while designing their strategies for such investments rather than minimising risks.

The research also contributes in providing useful information to help the developing countries such as Malaysia, Indonesia and Thailand in achieving the sustainable development Goals (SDGs) which also known as global goals. The

knowledge and results achieved by this research are useful in achieving the seventh goal—"Affordable and Clean Energy", twelfth goal—"Responsible Consumption and Production" and thirteenth goal—"Climate Action". This is because increasing in consumptions and investments in renewable energy resources are one of the most important criteria if the countries are to achieve the goals by 2030. By understanding the factors that influence individual investors' intention to invest in renewable energy investments, the government may take suitable actions such as introduce new incentive to increase the private investments. Consequently, this may increase individual investors' interest to invest in such investment. Indirectly, the country will be moving one step forward in achieving the SDGs.

Future research should consider the inclusion of non-investors' responses to obtain more generalizable results. In order to have a better address for variations in renewable energy investments' intention, future studies could also assess other construct variables such as moral norm, environmental consciousness and financial literacy that deemed to be useful in predicting investors' intention to invest in renewable energy investments.

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References

- Adam AA, Shauki ER (2014) Socially responsible investment in Malaysia: behavioral framework in evaluating investors' decision making process. *J Clean Prod* 80:224–240
- Aguilar FX, Cai Z (2010) Exploratory analysis of prospects for renewable energy private investment in the U.S. *Energy Econ* 32:1245–1252
- Ajadi, T., Cuming, V., Boyle, R., Strahan, D., Kimmel, M., & Logan, M. (2020). Global trends in renewable energy investment 2020. Frankfurt School – UNEP Collaborating Centre for Climate & Sustainable Energy Finance.
- Ajzen I (1991) The theory of planned behavior. *Organ Behav Hum Decis Process* 50(2):179–211
- Ajzen I, Fishbein M (1980) Understanding attitudes and predicting social behavior. Prentice-Hall, Englewood Cliffs
- Akhtar F, Das N (2018) Predictors of investment intention in Indian stock markets: extending the theory of planned behaviour. *Int J Bank Mark* 37(1):97–119
- Al Balushi Y, Locke S, Boulanour Z (2018) Islamic financial decision-making among SMEs in the Sultanate of Oman: aAn adaption of the theory of planned behaviour. *J Behav Exp Financ* 20:30–38
- Al Mamun A, Mohamad MR, Yaacob MR-B, Mohiuddin M (2018) Intention and behavior towards green consumption among low-income households. *J Environ Manag* 227:73–86
- Alleyne P, Broome T (2011) Using the theory of planned behaviour and risk propensity to measure investment intentions among future Investors. *J East Caribb Stud* 36(1):1–20
- Aren S, Zengin AN (2016) Influence of financial literacy and risk perception on choice of investment. *Procedia Soc Behav Sci* 235:656–663
- Arslan M, Şar S (2018) Examination of environmentally friendly “green” logistics behavior of managers in the pharmaceutical sector using the theory of planned behavior. *Res Soc Adm Pharm* 14(11):1007–1014
- Ballester E, Bravo M, Pérez-Gladish B, Arenas-Parra M, Plà-Santamaria D (2012) Socially responsible investment: a multicriteria approach to portfolio selection combining ethical and financial objectives. *Eur J Oper Res* 216(2):487–494
- Bank Negara Malaysia (2013) Financial Services Act. 2013. Retrieved September 4, 2019, accessed from <https://www.bnm.gov.my/>
- Bauwens T, Devine-Wright P (2018) Positive energies? An empirical study of community energy participation and attitudes to renewable energy. *Energy Policy* 118:612–625
- Belgiawan PF, Schmöcker J-D, Abou-Zeid M, Walker J, Fujii S (2017) Modelling social norms: case study of students' car purchase intentions. *Travel Behav Soc* 7:12–25
- Boute A (2020) Regulatory stability and renewable energy investment: the case of Kazakhstan. *Renew Sust Energy Rev* 121:109673
- Chaiyapa W, Esteban M, Kameyama Y (2018) Why go green? Discourse analysis of motivations for Thailand's oil and gas companies to invest in renewable energy. *Energy Policy* 120:448–459
- Chen CF, Xu XJ, Frey S (2016) Who wants solar water heaters and alternative fuel vehicles? Assessing social-psychological predictors of adoption intention and policy support in China. *Energy Res Soc Sci* 15:1–11
- Chen W, Chen J, Ma YK (2020) Renewable energy investment and carbon emissions under cap-and-trade mechanisms. *J Cleaner Prod* 278:123341
- Choi D, Kim KJ (2019) Influences of environmental and hedonic motivations on intention to purchase green products: an extension of the theory of planned behavior. *Sustain Prod Consum* 18:145–155
- Cohen, J. (1988). Statistical power analysis for the behavioural sciences (2nd ed.). Hillsdale, New jersey: Lawrence Erlbaum Associates.

- Curtin J, McInerney C, Gallachóir BÓ, Salm S (2019) Energizing local communities—what motivates Irish citizens to invest in distributed renewables? *Energy Res Soc Sci* 48:177–188
- Di Falco S, Sharma S (2018) Investing in climate change adaptation: motivations and green incentives in the Fiji Islands. *Ecol Econ* 154:394–408
- Douglas M, Wildavsky A (1983) Risk and culture: an essay on the selection of technological and environmental dangers. University of California Press, Berkeley
- Drivas K, Rozakis S, Xesfingi S (2019) The effect of house energy efficiency programs on the extensive and intensive margin of lower-income households' investment behavior. *Energy Policy* 128:607–615
- Edwards ML (2018) Public perceptions of energy policies: predicting support, opposition, and nonsubstantive responses. *Energy Policy* 117:348–357
- Egli F (2020) Renewable energy investment risk: an investigation of changes over time. *Energy Policy* 140:111428
- Eryilmaz D, Homans FR (2016) How does uncertainty in renewable energy policy affect decisions to invest in wind energy? *Electr J* 29(3):64–71
- Fan B, Yang W, Shen X (2019) A comparison study of 'motivation-intention-behavior' model on household solid waste sorting in China and Singapore. *J Clean Prod* 211:442–454
- Fishbein M, Ajzen I (1975) Belief, attitude, intention, and behavior: an introduction to theory and research. Addison-Wesley
- Fornell C, Larcker DF (1981) Evaluating structural equation models with unobservable variables and measurement error. *J Mark Res* 18(1): 39–50
- Frey UJ, Klein M, Deissenroth M (2019) Modelling complex investment decisions in Germany for renewables with different machine learning algorithms. *Environ Model Softw* 118:61–75
- Gamel J, Menrad K, Decker T (2016) Is it really all about the return on investment? Exploring private wind energy investors' preferences. *Energy Res Soc Sci* 14:22–32
- Gamel J, Menrad K, Decker T (2017) Which factors influence retail investors' attitudes towards investments in renewable energies? *Sustain Prod Consum* 12:90–103
- Gerrans P, Moulang C, Feng J, Strydom M (2018) Individual and peer effects in retirement savings investment choices. *Pac Basin Financ J* 47:150–165
- Gold HA, Malhotra A, Segars AH (2001) Knowledge management: an organizational capabilities perspective. *J Manag Inf Syst* 1:185–214
- Grilli G, Notaro S (2019) Exploring the influence of an extended theory of planned behaviour on preferences and willingness to pay for participatory natural resources management. *J Environ Manag* 232:902–909
- Hair JF, Hult GT-M, Ringle CM, Sarstedt M (2017) A primer on partial least squares structural equation modeling (PLS-SEM), 2nd edn. SAGE Publications, Inc., Los Angeles
- Halder P, Pietarinen J, Havu-Nuutinen S, Pollanen S, Pelkonen P (2016) The theory of planned behavior model and students' intentions to use bioenergy: a cross-cultural perspective. *Renew Energy* 89:627–635
- He Y, Liao N, Bi J, Guo L (2019) Investment decision-making optimization of energy efficiency retrofit measures in multiple buildings under financing budgetary restraint. *J Clean Prod* 215:1078–1094
- Henseler J, Ringle CM, Sarstedt M (2015) A new criterion for assessing discriminant validity in variance-based structural equation modeling. *J Acad Mark Sci* 43(1):115–135
- Hu J, Harmsen R, Crijns-Graus W, Worrell E (2018) Barriers to investment in utility-scale variable renewable electricity (VRE) generation projects. *Renew Energy* 121:730–744
- International Energy Agency (2010) The economics of transition in the power sector. International Energy Agency, Paris
- Ji X, Li G, Wang ZH (2017) Impact of emission regulation policies on Chinese power firms' reusable environmental investments and sustainable operations. *Energy Policy* 108:163–177
- Jiang X, Ding ZH, Li XP, Sun J, Jiang YL, Liu R, ... Sun WB (2020). How cultural values and anticipated guilt matter in Chinese residents' intention of low carbon consuming behavior. *J Cleaner Prod* 119069.
- Judge M, Warren-Myers G, Paladino A (2019) Using the theory of planned behaviour to predict intentions to purchase sustainable housing. *J Clean Prod* 215:259–267
- Kaffashi S, Shamsudin MN (2019) Transforming to a low carbon society; an extended theory of planned behaviour of Malaysian citizens. *J Clean Prod* 235:1255–1264
- Klein M, Deissenroth M (2017) When do households invest in solar photovoltaics? An application of prospect theory. *Energy Policy* 109:270–278
- Kline RB (2011) Principles and practice of structural equation modeling. Guilford Press, New York
- Li C, Shi Y, Wu C, Wu Z, Zheng L (2016) Policies of promoting entrepreneurship and angel investment: evidence from China. *Emerg Mark Rev* 29:154–167
- Li CH, Xu Y, Gill A, Haider ZA, Wang Y (2018a) Religious beliefs, socially responsible investment, and cost of debt: evidence from entrepreneurial firms in India. *Emerg Mark Rev* 38:102–114
- Li Q, Long R, Chen H (2018b) Differences and influencing factors for Chinese urban resident willingness to pay for green housings: evidence from five first-tier cities in China. *Appl Energy* 229:299–313
- Liao C, Zhao D, Zhang S (2018) Psychological and conditional factors influencing staff's takeaway waste separation intention: an application of the extended theory of planned behavior. *Sustain Cities Soc* 41:186–194
- Lim TS, Mail R, Karim M-RA, Ulum ZK-A-B, Jaidi J, Noordin R (2018) A serial mediation model of financial knowledge on the intention to invest: the central role of risk perception and attitude. *J Behav Exp Financ* 20:74–79
- Liu W, Zhang X, Feng S (2019) Does renewable energy policy work? Evidence from a panel data analysis. *Renew Energy* 135:635–642
- Ma J, Hipel KW, Hanson ML, Cai X, Liu Y (2018) An analysis of influencing factors on municipal solid waste source-separated collection behavior in Guilin, China by using the theory of planned behavior. *Sustain Cities Soc* 37:336–343
- Mauder KL, Park B, Judd CM (1996) Stereotypes, prejudice, and judgments of group members. *J Exp Soc Psychol* 32:411–436
- Moon WK, Kahlor LA, Olson HC (2020) Understanding public support for carbon capture and storage policy: the roles of social capital, stakeholder perceptions, and perceived risk/benefit of technology. *Energy Policy* 138:111312
- Nielsen E, Jolink A, Chappin M (2018) Investments in the Dutch onshore wind energy industry: a review of investor profiles and the impact of renewable energy subsidies. *Renew Sust Energy Rev* 81:2519–2525
- Obeng-Darko NA (2019) Why Ghana will not achieve its renewable energy target for electricity. Policy, legal and regulatory implications. *Energy Policy* 128:75–83
- Ojo AO, Raman M, Downe AG (2019) Toward green computing practices: a Malaysian study of green belief and attitude among Information Technology professionals. *J Clean Prod* 224:246–255
- Ozorhon B, Batmaz A, Caglayan S (2018) Generating a framework to facilitate decision making in renewable energy investments. *Renew Sust Energy Rev* 95:217–226
- Palamida E, Papagiannidis S, Xanthopoulou D (2018) Linking young individuals' capital to investment intentions: comparing two cultural backgrounds. *Eur Manag J* 36(3):392–407
- Park E, Ohm JY (2014) Factors influencing the public intention to use renewable energy technologies in South Korea: effects of the Fukushima nuclear accident. *Energy Policy* 65:198–211

- Paul J, Modi A, Patel J (2016) Predicting green product consumption using theory of planned behavior and reasoned action. *J Retail Consum Serv* 29:123–134
- Phan KC, Zhou J (2014) Factors influencing individual investor behavior: an empirical study of the Vietnamese stock market. *Am J Bus Manag* 3(2):77–94
- Qi X, Ploeger A (2019) Explaining consumers' intentions towards purchasing green food in Qingdao, China: the amendment and extension of the theory of planned behavior. *Appetite* 133:414–422
- Romano AA, Scandurra G, Carfora A, Fodor M (2017) Renewable investments: the impact of green policies in developing and developed countries. *Renew Sust Energ Rev* 68:738–747
- Salem R (2019) Examining the investment behavior of Arab women in the stock market. *J Behav Exp Financ* 22:151–160
- Salm S, Hille SL, Wüstenhagen R (2016) What are retail investors' risk-return preferences towards renewable energy projects? A choice experiment in Germany. *Energy Policy* 97:310–320
- Saunila M, Ukko J, Tero R (2018) Sustainability as a driver of green innovation investment and exploitation. *J Clean Prod* 179:631–641
- Sivaramakrishnan S, Srivastava M, Rastogi A (2017) Attitudinal factors, financial literacy, and stock market participation. *Int J Bank Mark* 35(5):818–841
- Sreen N, Purbey S, Sadarangani P (2018) Impact of culture, behavior and gender on green purchase intention. *J Retail Consum Serv* 41:177–189
- Stanley M (2016). Sustainable signals: the asset manager perspective. Instit Sustain Invest Bloomberg L.P.
- Tabachnick BG, Fidell SL (2013) Using multivariate statistics, 6th edn. Pearson Education, New Jersey
- Taghizadeh-Hesary F, Yoshino N (2019) The way to induce private participation in green finance and investment. *Financ Res Lett* 31:98–103
- Tan CS, Ooi HY, Goh YN (2017) A moral extension of the theory of planned behavior to predict consumers' purchase intention for energy-efficient household appliances in Malaysia. *Energy Policy* 107:459–471
- Toni M, Renzi MF, Mattia G (2018) Understanding the link between collaborative economy and sustainable behaviour: an empirical investigation. *J Clean Prod* 172:4467–4477
- Trivedi RH, Patel JD, Acharya N (2018) Causality analysis of media influence on environmental attitude, intention and behaviors leading to green purchasing. *J Clean Prod* 196:11–22
- Uche CU (2000) The theory of regulation: a review article. *J Financ Regul Compliance* 9(1):67–80
- Verma VK, Chandra B (2018) An application of theory of planned behavior to predict young Indian consumers' green hotel visit intention. *J Clean Prod* 172:1152–1162
- Wan C, Shen GQ (2013) Perceived policy effectiveness and recycling behaviour: the missing link. *Waste Manag* 33:783–784
- Wan C, Shen GQ, Yu A (2014) The role of perceived effectiveness of policy measures in predicting recycling behaviour in Hong Kong. *Resour Conserv Recycl* 83:141–151
- Wan C, Shen GQ, Choi S (2018) Understanding public support for recycling policy: to unveil the political side of influence and implications. *Environ Sci Policy* 82:30–43
- Wang C, Zhang J, Yu P, Hu H (2018) The theory of planned behavior as a model for understanding tourists' responsible environmental behaviors: the moderating role of environmental interpretations. *J Clean Prod* 194:425–434
- Wang QC, Chang RD, Xu Q, Liu X, Izzy YJ, Ma YT, Wang YX (2021) The impact of personality traits on household energy conservation behavioral intentions – an empirical study based on theory of planned behavior in Xi'an. *Sustain Energy Technol Assess* 43: 100949
- Wang, Y., Liang, J., Yang, J., Ma, X., Li, X., Wu, J., .Feng, Y. (2019). Analysis of the environmental behavior of farmers for non-point source pollution control and management: an integration of the theory of planned behavior and the protection motivation theory. *J Environ Manag*, 237, 15–23.
- Warsame MH, Ireri EM (2016) Does the theory of planned behaviour (TPB) matter in Sukuk investment decisions? *J Behav Exp Financ* 12:93–100
- Wilson C, Pettifor H, Chrysoschoidis G (2018) Quantitative modelling of why and how homeowners decide to renovate energy efficiently. *Appl Energy* 212:1333–1344
- Wu T, Yang SW, Tan JJ (2020) Impacts of government R&D subsidies on venture capital and renewable energy investment – an empirical study in China. *Res Policy* 68:101715
- Wunder Capital. (2017). Why individual investors matter in the transition to a clean energy economy. Retrieved October 17, 2020, from Wunder Capital: https://blog.wundercapital.com/2017/07/05/why-individual-investors-matter-in-the-transition-to-a-clean-energy-economy/?fbclid=IwAR2pmIY_n3L_WIHIdunGpCEhiu986nfXcFC6em54hsJne7vwBi125F_NmTY. Accessed 3 Mar 2019
- Yadav R, Pathak GS (2016) Young consumers' intention towards buying green products in a developing nation: extending the theory of planned behavior. *J Clean Prod* 135:732–739
- Yadav R, Pathak GS (2017) Determinants of consumers' green purchase behavior in a developing nation: applying and extending the theory of planned behavior. *Ecol Econ* 134:114–122
- Zhang CY, Yu B, Wang JW, Wei YM (2018) Impact factors of household energy-saving behavior: an empirical study of Shandong Province in China. *J Clean Prod* 185:285–298

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